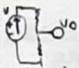
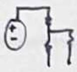
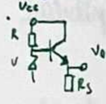
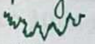
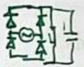
Teorie

- $C_{paralel} = C_1 + C_2$
- 

$\Rightarrow V_{out} = V_1 \cdot \frac{R_2}{R_1+R_2}$
- $i_C = \beta \cdot i_B$
- $V_{in} = V_{senzor} \cdot \frac{Z_{in}}{Z_{in}+Z_{senzor}}$
- $i_R = \frac{V}{R}$
- 7905 stabil. tens. ⇒ -5V
- stab. liniar tens.
- Val. căderi tensiune:
$-V_{SAN_1} \cdot \frac{R_2}{R_1} + V_{SON_2} \cdot \left(1+\frac{R_2}{R_1}\right)$
- $i_R = \frac{V_1 - (V_{D1}+V_{D2})}{R_1}$
- 

$E_{Th} = E \cdot \frac{R_2}{R_1+R_2}$
$I_{sc} = \frac{V_1}{E_{Th}}$ $\Rightarrow V = E_{Th} \cdot \frac{R_3}{R_{Th}+R_3}$
$R_{Th} = \frac{R_1 \cdot R_2}{R_1+R_2}$
- Tensiune ideală are impedanța 0
- $I_{d,min} = \frac{V_1 - (U_{D1}+U_{D2})}{R_1}$
- pt. U_{CE}: $U_E = U_B - U_{BE}$
$U_B = V_{CC} \cdot \frac{R_{b2}}{R_{b2}+R_{b1}}$
$i_E = \frac{U}{R} = i_C$; $U_{RC} = i_C R_C$
$U_{CE} = V_{CC} - U_{RC} - U_{RE}$
-

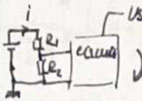
RAN
BL
- $i_D = \frac{E - U_d}{R}$
- 

$V_{out} = U_B - U_{BE}$
- $V_1 = -\frac{R_2}{R_3}$ pt. RS
- Un filtru rejectează banda de ord. 2, activ, topologie Sallen-Key
- Filtru trece sus: Vin 
vout mic
-  punte redresoare
- FTJ de ord. 2: -40dB
- Buck converter
- căd. tens. LED: 1,6 și 4 V
- Dioda Schottky: metal-semiconductor
- Repetor - tensiune: colector comun
- amplificator tensiune ideal: At f. mare
i in f. mare
i out f. mică
- AO NU are amplificarea în buclă deschisă de val. <.
- CMMR: < fact. rejecție modului comun
$20\log_{10}\left(\frac{A_d}{A_c}\right) \simeq 0{,}01$
- transimpedanță: curent → tensiune
- montaj emitor comun: Sin defazat de Sout
- AO NU se folosește con. buclă închisă cu reacție pozitivă
- AO ideal: curenții de intrare 0
- B pt. TBIP: 20-300
- capacitatea bateriilor se măsoară în Ah
- AO tipic: zeci de microV
- Limitează tensiunea pe in. AO-ului de Vad
- $I = \frac{V}{R_5}$ cu V = LM78⑫ ⇒ V = 12V
- $E_C = 5V \Rightarrow D_1$ deschis; D_2 închis
- 20dB/decadă
- -50% ⇒ 25%
-

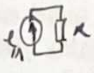
$U_{ab} = b \Rightarrow c = a - b$
[C/R] interval
- 5%·5000 = 250; între 4750 și 5250
- $Z_{in} = \frac{Z_0 \cdot U_0'}{U_0 - U_0'} = 99\,k\Omega$
- $V_{i+} = U_{i-}$
$V_{i+} = U_i \Leftrightarrow U_i = I_0 \cdot R_0$
$\Rightarrow I_0 = \frac{U_i}{R_0} = 10\,mA$
valoarea lui I_0

1. Noțiuni fundamentale:

- Curentul electric: mișcarea ordonată a purtătorilor de sarcină: $i = \frac{\Delta Q}{\Delta t}$; Amperi
- Tensiunea electrică: $V_{AB} = V_A - V_B$; Volt
- Energie electrică: Jouli; $W = P \cdot t$
- Rezistența: $R \cdot I = U$; Ohm
- $R_{es} = R_1 + R_2$; $\frac{1}{R_{ep}} = \frac{1}{R_1} + \frac{1}{R_2}$
- $\frac{1}{C_{es}} = \frac{1}{C_1} + \frac{1}{C_2}$; $C_{ep} = C_1 + C_2$
- Bobine: ca la R
- Divizorul de tensiune
$U_0 = \frac{R_2}{R_1+R_2} \cdot E$

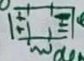


echivalare: $e = \frac{R_2}{R_1+R_2} \cdot E$
$r = \frac{R_1 R_2}{R_1+R_2}$

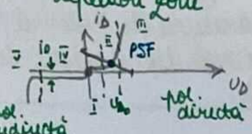


- Thevenin: $R_{Th} = \frac{E_{Th}}{I_{sc}}$
- Puterea: $P = I \cdot V$

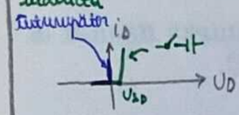
2. Dioda semiconductoare:

- gol ⇒ G = 0; fir ⇒ R = 0
- $[G]_{S.I} = MHO / S$
- limitarea curentului / măsurare
- Capacitate: $i_C = \frac{dQ_C}{dt}$ ($\Leftrightarrow Q_C = C V_C$); $i_C = C\frac{dv_C}{dt}$
- Inductanța: $i_C = C\frac{dv_C}{dt}$; $v_L = L\frac{di}{dt}$; $Z_L = j\omega L$
reactivă
- el. dipol; permite trecerea curentului într-un singur sens
-  electropozitivă
depletion zone

$$i_D(U_D) = I_0\left(e^{\frac{qU_D}{kT}} - 1\right)$$

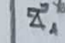


I - zona de tensiuni și curenți mici la pd.
II - zona diodei reale, exponențială
III - zona curenților mari; zona liniară
IV - zona curenților mici la pi.
V - zona de străpungere
$U_D \cong U_{str}$

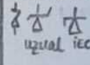


Diode redresoare: AC-DC; uz general între 50Hz, 400Hz

Diode Schottky: joncțiune metal-semiconductor
tensiune între 0,3-0,4V.
tensiune inversă zeci de N μA/nA
foarte rapide



Diode Zener: U_S bine definită; stabilizatoare
U limit; tensiune de străpungere 5%
< tunelare cuantică (<5V)
izolare în avalanșă (>5V)

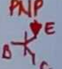


Led: Light Emitting Diode
lumină monocromatică; proporțional cu DC
1V - infraroșu; 3V - albastru

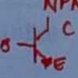
Fotodiode: curent invers; i_0 prop. cu iluminarea
generare perechi electron-gol; receptoare
fibră optică; panouri fotovoltaice

3. Tranzistorul bipolar:

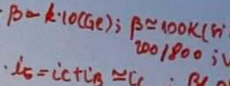
- Rezistență a cărei valoare poate fi modificată
- Lucrăm cu semiconductoare
- 2 joncțiuni, una mai puternică decât alta: lucrăm ori cu PNP/NPN
- BJT și TBIP în polarizare directă

PNP


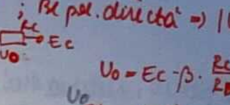
KCL: $i_E = i_C + i_B$
KVL: $U_{EC} = U_{EB} + U_{BC}$

NPN


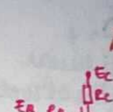
KCL: $i_E = i_C + i_B$
KVL: $U_{CE} = U_{CB} + U_{BE}$
$i_C = \beta \cdot i_B$

- $\beta \sim k \cdot 10$ (Ge); $\beta \simeq 100K$ (Si); $\beta = 10000K$ (tranzistoare compuse)
200/800; variază
- $i_E = i_C + i_B \simeq i_C$; BE pol. directă ⇒ $|U_{BE}| \geq |U_{BE0}|$
- 

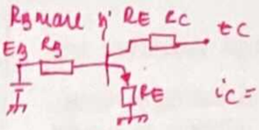
$U_0 = E_C - \beta \cdot \frac{R_C}{R_B}(U_i - U_{BE})$



BL: $i_C = 0$ ⇔

EM și RB


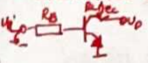
$i_C = \frac{\beta(E_B - U_{BE})}{R_B}$

RB mare și RE


$i_C = \frac{E_B - U_{BE}}{\frac{R_B}{\beta} + R_E}$

$R_E \gg R_B \Leftrightarrow \frac{R_{B1}}{\beta}$ neglijabil

a) circuitul cu emitor legat la masă



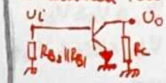
$U_0 = E_C - i_C R_C$
$i_C = \beta i_B$
$i_B = \frac{U_i - U_{BE}}{R_B}$

$U_i = i_B R_B + U_{BE}$ $\begin{cases} U_i \leq U_{BE0} \Rightarrow i_B = 0 \\ U_i \geq U_{BE0} \Rightarrow U_{BE} = ct. = U_{BE0} \end{cases}$

- Caracteristici: de intrare (in(in))
de ieșire (out(out))
de transfer (out(in))
- Conexiuni posibile: EC/CC/BC
- Ec. hibride: $\begin{cases} U_i = h_i i_i + h_r u_0 \\ i_0 = h_f i_i + h_o u_0 \end{cases}$

h_i = impedanță
$h_f = h_r$ = adim.
h_o = admitanță Ω^{-1} [S/MHO]

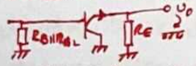
- Conexiunea cu Emitor comun



- Zi mică; ii mare
- Au, Ai mari
- Zo ~ medie

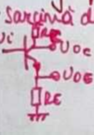
$Z_i \approx h_i$; $Z_0 \approx R_C$, $A_u = -\frac{h_f}{h_i} \cdot R_C$; $A_i \approx -h_f$

- Conexiune colector comun



- $A_u \approx 1$
- $A_i \approx h_f$
- $Z_0 = R_E \| \frac{h_i}{h_f}$
- Zi f. mare
$Z_i = h_i + h_f \cdot R_E$

- montaj repetor pe emitor
- Sarcină distribuită:



$U_{OC} = -U_{OE} \cdot \frac{Z_C}{Z_E}$
$U_{OE} \approx U_i$; $i_C \approx i_E$

	EC	CC	SD
Au	mare dep. de hf, hi	1	precisă
Zi	mică hi	mare	mare
Zo	mare	mică	C - mare E - mică
Obs	amplifică și inversează	repetă	ieșiri din antifază
Aplicații	et. intermed	et. in/out	ampl. simplu

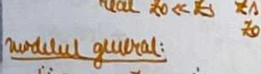
4. Amplificatoare

- baza AO
- caract. in: $Z_i = \frac{U_i}{i_i}$
- caract. out: $Z_0 = \frac{U_{0\infty} - U_0}{i_0}$
- amplificarea în curent: $A_i = \frac{i_0}{i_i}$
- amplificarea în tensiune: $A_U = \frac{U_0}{U_i}$

Intrare: ideal $U_i = U_g \Leftrightarrow Z_i \to +\infty$
real: $Z_i \gg Z_g$ $Z_g = 1K$
$Z_i = 1M$

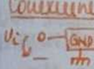
Ieșire: $U_0 = U_{0\infty} \cdot \frac{Z_s}{Z_0 + Z_s}$
ideal $U_0 = U_{0\infty} \Rightarrow Z_0 \to 0$
real $Z_0 \ll Z_s$ $Z_s = 1K$
$Z_0 = 10K$

modelul general:



Conexiune single-ended

Conexiune diferențială

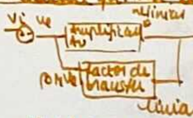


$V_0 = A_d(V_{i+} - V_{i-})$
$U_{0d} = U_{0+} - U_{0-}$
$V_{id} = V_{i+} - V_{i-}$

- Vid - tensiunea de intrare dif.
$V_{id} = V_{i1} - V_{i2}$
- Vic - mod comun de intrare
$V_{ic} = \frac{V_{i1}+V_{i2}}{2}$ (zgomotul)
- $V_0 = A_d(V_{i1} - V_{i2}) + A_c\left(\frac{V_{i1}+V_{i2}}{2}\right)$
- $CMRR = 20\lg\frac{A_d}{A_c}$;
- $PSRR = 20\lg\frac{\Delta V_{alim} A_d}{\Delta V_0}$;
- Ideal: $A_d \to 0$; $CMRR \to \infty$; $PSRR \to +\infty$

5. AO și aplicații

- intrare de tip single-ended (open loop system).
- montaje în buclă închisă de reacție
- feedback: ieșirea se întoarce la intrare
- ajută montajul să vadă ce se întâmplă la ieșire; ușor liniarizate
- modelul pt. rist. de reacție:



$V_o(V_i)$ = liniar

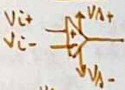
$V_o = V_i \cdot \frac{1}{\frac{1}{A_u} + \beta_r}$

- ideal $A_u \to +\infty$

AO

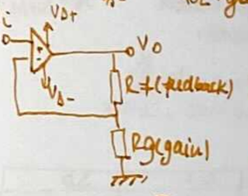
- componentă electronică; amplifică diferența $V_{i+} - V_{i-}$
- ideal $Z_i \to \infty$; $Z_i \sim 100 M\Omega$
- curenții de intrare i_{i+}, i_{i-} f. mici
- $Z_o \sim 100 \Omega$
- $A_u = 10^5$: amplificarea de tensiune
- Rail to Rail Output $V_{D-} \le V_o \le V_{D+}$
- Rail to Rail Input $V_{D-} \le V_{ic} \le V_{D+}$
- Rail to Rail Input Output

AO neinversor; buclă deschisă ⊕



$V_o = A_{uo}(V_{i+} - V_{i-}) = A_{OL}(V_{i+} - V_{i-})$

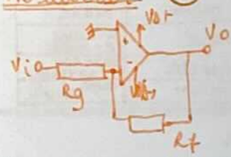
$A_{OL} \to$ gain (open-loop)



$V_o = \frac{V_i}{\frac{1}{A_{OL}} + \frac{R_g}{R_f + R_g}}$

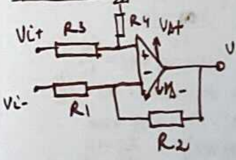
$V_o = V_i\left(1 + \frac{R_f}{R_g}\right)$

AO inversor ⊖



$V_o = -V_i \cdot \frac{R_f}{R_g}$

AO inversor - neinversor

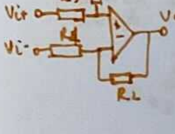


$V_o = -\frac{R_2}{R_1} \cdot V_{i-}$

$V_o = V_{i+} \cdot \frac{R_4}{R_3 + R_4}\left(1 + \frac{R_2}{R_1}\right) - \frac{R_2}{R_1} V_{i-}$

$V_o = \frac{R_2}{R_1}(V_{i+} - V_{i-})$

AO diferențial

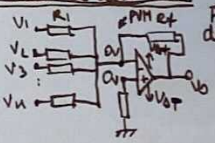


$V_o = \frac{R_2}{R_1}(V_{i+} - V_{i-}) + V_{OFFSET}$

V_{OFFSET} - modifică poziția semnalului

$GBP = A_u \cdot f$

Sumatorul inversor



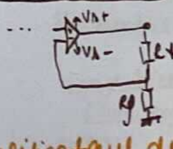
PVM: punct virtual de masă; NU GND

$i = i_1 + \dots + i_n$

$V_o = -\frac{R_f}{R}(V_1 + \dots + V_n)$

$R_1 = R_2 = \dots = R_n$

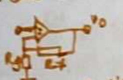
Sumator neinversor



$V_o = \frac{V_1 + \dots + V_n}{n}\left(1 + \frac{R_f}{R_g}\right)$

Amplificatorul de instrumentație

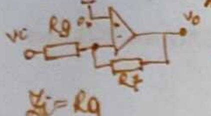
- single-ended



Z_i f. mare

$V_o = \left(1 + \frac{R_f}{R_g}\right) V_i$

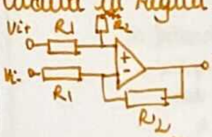
ainversor



$\Rightarrow V_o = -\frac{R_f}{R_g} \cdot V_i$

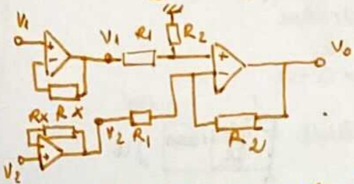
$Z_i = R_g$

- Circuite în regim diferențial



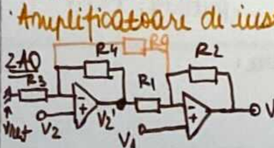
$V_o = (V_{i+} - V_{i-}) \cdot \frac{R_2}{R_1}$

amp. dif $\Leftrightarrow \frac{R_2}{R_1} = \frac{R_4}{R_3}$



$V_o = \frac{R_2}{R_1}(V_1 - V_2)$

- un AO poate lucra ca repetor în tensiune ⇔ "unity gain stable", adică stabil la amplificarea unitate
- Amplificatoare de instrumentație: 2 AO / 3 AO



$V_o = \left(1 + \frac{R_2}{R_1}\right)(V_1 - V_2)$

V_2 = impedanța

$V_2' = V_2\left(1 + \frac{R_4}{R_3}\right)$ ampl. neinversor

- cu câștig reglabil: $V_o = (V_1 - V_2)\left(1 + \frac{R_2}{R_1} + \frac{2R_2}{R_g}\right) + V_{ref}$

3AO

- dacă INA cu 2AO nu au rezistențele potrivite ⇒ CMMR

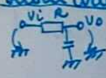
$V_o = (V_1 - V_2) \cdot \frac{R_2}{R_1}\left(1 + \frac{2R_3}{R_g}\right)$; $Gain = A_u = \frac{R_2}{R_1}\left(1 + \frac{2R_3}{R_g}\right)$

- aplicații practice: aparatura de măsură / medicală (Z_i f. mare; măs. dif. și tensiuni de amplitudine mică)

6. Filtre

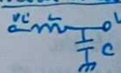
- **PASIVE**: funcția de transfer depinde de componentele pasive (R, L, C).

Filtru trece JOS RC PASIV



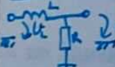
- filtru de ordin 1
- H scade cu -20 dB/decadă
- $H = \frac{1}{1 + j\frac{\omega}{\omega_0}}$ cu $\omega_0 = \frac{1}{RC}$
- dacă înlocuim R cu C ⇒ FTS

Filtru trece JOS PASIV LC



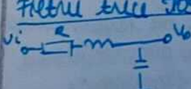
- $H = \frac{1}{1 - \frac{\omega^2}{\omega_0^2}}$ cu $\omega_0 = \frac{1}{\sqrt{LC}}$
- ordin 2; $H \to \infty \Leftrightarrow \omega_0 \to \omega$

Filtru trece JOS PASIV RL



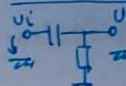
- $U_o = \frac{R}{R + Z_L} U_i$

Filtru trece JOS PASIV RLC



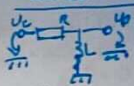
- $H = \frac{1}{1 + 2j\xi\frac{\omega}{\omega_0} - \frac{\omega^2}{\omega_0^2}}$
- $\xi = \frac{1}{2}RC\omega_0 = \frac{R}{2}\sqrt{\frac{C}{L}}$; $\omega_0 = \frac{1}{\sqrt{LC}}$
- factor de calitate $Q = \frac{1}{R}\sqrt{\frac{L}{C}}$

Filtru trece SUS RC



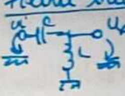
- ordin 1

Filtru trece SUS RL

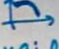


- ordin 1

Filtru trece SUS LC



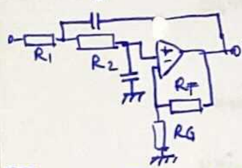
- ordin 2

- **ACTIVE**
  - det. ordin filtru: puterea lui ω; -20·n dB/dec pt. ordinul n
  - tipuri: - Butterworth 
    - Bessel: defazaj liniar
    - Cebâșev: tranziție abruptă
  - tipuri implementări

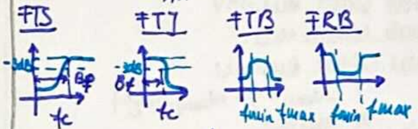
· PASIVE	· ACTIVE
RC, RL, LC, RLC	Sallen-Key, cu reacție multiplă, cu var. de stare, cu conv. de Z

- DIGITALE (NUMERICE): hardware / software, răspuns finit / infinit la impuls

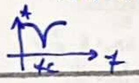
Sallen-Key:



- amplificare $k = 1 + \frac{R_f}{R_G}$
- stabil $k < 3$

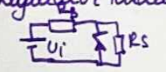


Filtru Notch



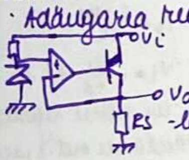
7. Surse de tensiune

- Stabilizator parametric: element regulator paralel



$U_o = V_Z = V_{Z0} + (U_i - U_o)\frac{r_Z}{R_S} - \frac{U_o \cdot r_Z}{R_S}$

- $\eta = \frac{U_{util}}{U_{util} + P_{iedere}}$
- sursele au η ce depinde de V_o și U_i, dar nu de implementare
- Adăugarea reacției

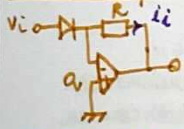


montaj cap-tabular

- LDO → tensiunea care cade pe regulator mai mică ≈ 0,1 V ($U_i - U_o$)

Circuite neliniare cu AO

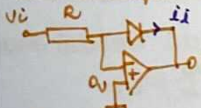
Convertor exponențial: $V_i > 0$



$V_o = -i_i R = -R i_o e^{\frac{V_i}{V_T}}$

Convertor logaritmic: $V_i > 0$

- dacă $V_i > 0$ nu se respectă, dioda este blocată și nu mai e reacție negativă ⇒ devine comparator



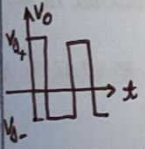
$V_o = -V_T \ln\frac{i_i}{I_o} = -V_T \ln\frac{V_i}{R I_o}$

Multiplicator

- ridicarea la pătrat, radical...
- $ab = e^{\ln a + \ln b}$

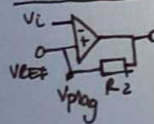
Comparatorul: buclă deschisă

- $V_o = A_{vo}(V_{i+} - V_{i-})$; V_o f. mare



- neidealități: zgomot V_i, V_{REF}
- zgomot mare ⇒ multe tranziții

CU HISTEREZIS (+)



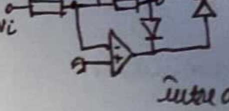
$V_{prag} = \frac{V_{REF} R_2 + V_o R_1}{R_1 + R_2}$

V_o: V_{D-} / V_{D+}

- se ignoră zgomotul

REDRESOR IDEAL MONOALTERNANȚĂ

- dioda ideală



- o diodă deschisă, una închisă
- dioda deschisă asigură reacția negativă
- comutarea între diode este rapidă